

FA26 NT@B Software Division Recruitment Project

Overview

This challenge is a compact version of the work our research teams do. There is no single correct solution and no target accuracy. We are interested in how you interrogate a problem, what you notice that you were not told to look for, and whether you can defend your reasoning under questioning.

Time budget: 5 to 10 hours. A thoughtful, complete submission at six hours beats an exhausted sprawling one at twenty. If you pass ten hours, stop and write up where you are.

Use AI tools freely. We assume you will, and we have designed this so that the parts that matter are the parts an AI cannot do for you. You will defend your choices in a live conversation afterward, so build things you understand.

Background

Motor imagery is imagining a movement without performing it. Imagining a left-hand versus right-hand movement modulates sensorimotor rhythms over the contralateral hemisphere, and scalp EEG picks up some of this. It is the basis of most non-invasive brain-computer interfaces. It is also weak, noisy, and unreliable, which is what makes it interesting.

Our teams build models that must work on data they were never trained on: a new person, a new recording session, a brain region nobody recorded from. Almost every design decision we make is downstream of that constraint. This challenge is a small version of that problem.

Dataset

PhysioNet EEG Motor Movement/Imagery Dataset (EEGMMIDB)

109 subjects, 64 channels, 160 Hz, recorded with the BCI2000 system.

- Dataset: <https://physionet.org/content/eegmmidb/1.0.0/>
- Original paper: Schalk et al., *IEEE Trans. Biomed. Eng.* 51(6):1034-1043 (2004). doi:10.1109/TBME.2004.827072
- PhysioNet resource paper: Goldberger et al., *Circulation* 101(23):e215-e220 (2000)

You obtain the data yourself. There is no bundled subset and no starter notebook. Getting the data into a workable form is part of the exercise.

Useful tooling, all optional:

- MNE-Python, which has a built-in fetcher:
`mne.datasets.eegbci.load_data(subject, runs)`
https://mne.tools/stable/generated/mne.datasets.eegbci.load_data.html
- MNE tutorials: https://mne.tools/stable/auto_tutorials/index.html
- MOABB, which wraps this dataset as **PhysionetMI** with standardized paradigms:
<https://moabb.neurotechx.com/docs/generated/moabb.datasets.PhysionetMI.html>
- Braindecode, for deep learning on EEG: <https://braindecode.org/>

Run structure. Each subject completed 14 runs. The ones relevant here:

Runs	Content
1, 2	Baseline, eyes open / eyes closed
3, 7, 11	Left vs. right fist, executed
4, 8, 12	Left vs. right fist, imagined
5, 9, 13	Both fists vs. both feet, executed
6, 10, 14	Both fists vs. both feet, imagined

Within a run, annotation T0 marks rest, T1 and T2 mark the two conditions. **The meaning of T1 and T2 depends on the run.** Check this yourself.

Scope. Use as many or as few subjects as you need. More subjects cost download time; fewer may limit what you can conclude. That is your call to make and to justify.

The Task

Build a classifier that distinguishes left fist from right fist.

You may use the executed runs, the imagined runs, or both. Imagery is the harder and more BCI-relevant problem; execution has a stronger signal. Working with both, or asking whether a model trained on one transfers to the other, is fair game and possibly interesting.

Any approach is fine: spectral features into a linear model, CSP, Riemannian methods, a convolutional network, a transformer, whatever you want to try. Ambition is welcome. So is a simple approach, provided you can say why it is the right one.

What We Are Actually Assessing

Assume we are skeptical readers. When you report a performance figure, our first question is not "is it high" but "what else could explain this number?" Your submission should answer that question before we ask it.

Specifically, your evaluation should establish:

Explain your data processing pipeline. How did you filter the dataset or process your data? How did you deal with noise?

What your model can and cannot do. A single accuracy figure describes one situation. Which situation? Under what conditions does the number hold, and under what conditions does it fall apart? Show us the range, not the best case.

That the result is not an artifact of how you measured it. Any evaluation makes assumptions about what the training and test data share. How did you decide what the train and test were?

What the appropriate baselines are. How much of your performance is attributable to something other than the thing you claim to be decoding? Chance is one reference point and a weak one. Consider looking into the dataset and the different subjects and understand why your model would be suggesting a certain output.

How much of this is about the person rather than the task. These recordings come from many individuals, and individuals differ from each other in ways that have nothing to do with what they were imagining. Quantify how much this matters for your result.

Whether your model has learned or memorized. Where does it overfit, to what, and how do you know?

We are deliberately not telling you which techniques to use for any of this. Working out what the right analyses are is the challenge.

What to Expect from the Numbers

Chance baseline is 50 percent. This task is hard, and we expect the figure you end up defending to be considerably lower than the first figure you see. That gap is not a failure. Explaining it is most of the assignment.

We do not rank submissions by accuracy. While the performance of your model is important, we do take into account the creativity of your model and the choices you made.

Deliverables

1. Code that runs, as a Github repo. Please make sure it is public.

2. A prediction script. A trained model plus a function or CLI taking a raw EDF path and returning predicted labels, with pinned dependencies and a one-line usage example. We will run it on subjects you did not use. Not to score you, but as a consistency check and as something specific to talk about.

3. A demo video. 2-3 minutes (3 minutes max).

Required content

Your evaluation and what it establishes. The set of results you ran, why you ran those and not others, and what a skeptical reader should conclude from them. Try to log your attempts to be able to explain your attempts.

One design decision. A specific choice, a real alternative you considered, and why you went the way you did.

Your weakest point. What do you trust least in your own work, and what would have to be true for your conclusion to be wrong?

Something we did not ask about. One thing you found in this dataset that no prompt above covers. Deliberately open.

What you would do next with more time and more compute.

AI use. Which tools, for what. Then pick one specific choice in your project and explain it in your own words.

What we care about is walking us through your project, explaining your data processing decisions, architectural choices, and your evaluation metrics.

Logistics

One office hour at **Monday 1-3 PM, Tuesday 8-10 PM, Thursday 8-10 PM, Strada Cafe** for questions about setup and scope. We will not answer questions about how to evaluate your model.

Submit by **Friday, 09/11 11:59 PM** to [this Google Form](#).